6TH GRADE MATH · UNIT 5: INEQUALITIES

Theme Park Rules

Use inequalities to check ride rules, budgets, and wait times at a theme park.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your class is visiting Thunder Valley Theme Park. Every ride has a rule: you must be at least a certain height, some rides have weight or age limits, and your group has a budget for food and games.

You will write, solve, and graph inequalities to decide who can ride what and to plan the best day possible.



Ride Rule

Thunder Coaster

height $h \geq 48$ in

Budget

$40 for food

and games

Learning Goals

- I can write an inequality for a rule.
- I can graph an inequality on a number line.
- I can solve one-step inequalities.

Project Timeline

When	What you will do
Day 1	Ride Rules
Day 2	Budget Check
Day 3	Plan Your Day
Final	Theme Park Day Plan + reflection

Part 1: Ride Rules

Day 1 · Use the Ride Rule Cards. Let h be height in inches.

Ride Rule Cards

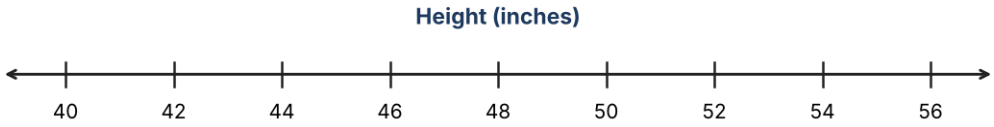
Thunder Coaster	Kiddie Cars	Sky Swings
Height at least 48 in	Height under 42 in	Height at least 44 in Weight at most 220 lb

Bumper Boats	Log Flume	Drop Tower
Age 8 or older	Height at least 46 in	Height at least 52 in

1. Write an inequality for the Thunder Coaster rule.

Starter: *h ___ 48*

2. Graph the Thunder Coaster rule on a number line. Use a closed circle or an open circle?



3. Write and graph the Kiddie Cars rule.

4. Maya is 47 inches tall. Which rides can she go on?

Part 2: Budget Check

Day 2 · Write an inequality and solve it.

5. Game tickets cost $3 each. You can spend at most $24. How many tickets t can you buy?

Starter: *3t ≤ 24, so t ≤ ___*

6. You have $40. After lunch you have to keep at least $15. Lunch costs x dollars. Write an inequality.

7. A ride takes 5 minutes. You have 45 minutes. How many times r can you ride?

8. Is 10 a solution of 3t ≤ 24? Explain.

Part 3: Plan Your Day

Day 3 · Plan a day for your group.

9. Pick three rides. List the height rule for each and say who in a group of heights 43, 46, 49, and 53 inches can ride.

10. Write an inequality for how much you can spend on snacks if snacks cost $4 each and your snack budget is $16.

11. Make a schedule for your day. Show that your total time on rides is at most 3 hours (180 minutes).

Final Product: Theme Park Day Plan

Your final product must include:

- Each ride rule as an inequality and a graph
- Which riders can go on each ride
- Your budget inequality and solution
- Your day schedule
- A poster of your favorite ride rule

Reflection

R1. When do you use a closed circle, and when do you use an open circle?

R2. Why does an inequality have many solutions?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Ride rules, budgets, and times are written, solved, and graphed as inequalities and used to make decisions. Goes beyond the minimum with a creative, realistic choice.	Ride rules, budgets, and times are written, solved, and graphed as inequalities and used to make decisions.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Day plan includes rules, graphs, solved inequalities, and a schedule. Polished, organized, and visually engaging.	Day plan includes rules, graphs, solved inequalities, and a schedule. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.